MIL-P-48367B (AR)
20 July 1982
SUPERSEDING
MIL-P-48367A (AR)
17 February 1977

MILITARY SPECIFICATION

PROPELLANT M30A1
FOR USE IN CHARGE PROPELLING M203

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M30A1 Propellant produced by batch process for use in the 155MM Propelling Charge, M203 (Zone 8) for the 155MM Howitzer, M198.

1.2 Classification. The propellant shall be Type I, cylindrical multi-perforated grains.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-C-48243 - Charge Propelling, 155MM, M203 Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

FSC: 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein."

DRAWINGS

U. S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9281897 - Charge, Prop, M203 for Howitzer, M198

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. The propellant shall be M30A1, conforming to the requirements of MIL-STD-652 and Dwg 9281897. The propellant shall contain 0.15% maximum graphite glaze (added).

3.2 Proving Ground assessments. The propellant when fired in the M199 cannon with 75% (minimum) remaining life and with the projectile, M549 (or PXR-6325) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.5.1. The contractor responsible for LAP of this charge in production shall manufacture all charges (reference, master calibration and test) in accordance with applicable requirements (see 6.6). The minimum LAP at one time for a reference or master calibration lot shall be 2000 charges.

3.2.1 Permissible individual maximum pressure. The propellant at recommended charge weight shall not produce a chamber pressure greater than 57,950 psi at any operating temperature (-65°F to +145°F) when fired in a standard weapon (see 6.11).

3.2.1.1 Differential pressure. The propellant shall not produce an individual differential pressure greater than 6,000 psi at 145°F or 3000 psi at -65°F.

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3.2.2 Charge establishment phase. At 70 degrees fahrenheit (^{o}F) and standard weapon conditions, the propellant shall be capable of assessment to a velocity of 2710 feet per second (fps).

3.2.3 Uniformity phase

3.2.3.1 Uniformity of velocity. The standard deviation of velocity for the lot at 70°F shall not exceed 7.0 fps.

3.2.3.2 Average pressure. At standard weapon conditions and recommended charge weight, the propellant shall not produce an average pressure greater than 47,500 pounds per square inch (psi), at 70°F and 55,000 pound per square inch (psi) at 145°F.

3.3 Preproduction inspection. This specification contains provisions for preproduction inspection. Requirements for the submission of preproduction inspection samples by the contractor shall be as specified in 4.3.1.

3.4 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all the applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to asure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The following types of inspection shall be conducted on the unit product:

a. Preproduction Sample Inspection
b. Quality Conformance Inspection

4.3 Preproduction sample (pilot lots). The preproduction sample is intended to provide the necessary information to establish the propellant granulation needed to meet the required ballistic parameters.

4.3.1 Submission. The Contractor shall select three (3) propellant granulations which, from experience, should best fit the requirements of this specification. The contractor shall submit a preproduction sample as designated by the Contracting Officer for evaluation in accordance with 4.3.2, consisting of three (3) pilot lots approximately 1200 lbs each representing the three (3) candidate granulations.

4.3.2 Inspections to be performed. Each of the pilot lots comprising the preproduction sample and submitted in accordance with 4.3.1 may be subjected by the Government to any or all of the requirements of this specification.

4.3.3 Rejection. Government Engineering and Product Assurance personnel and the Contractor shall review the pilot lot test data and shall select a granulation size which best fits the requirements of this specification. The Technical Agency shall notify the Contracting Officer of the selected granulation. The Contractor shall request approval from the Contracting Officer to produce the selected granulation. If none of the pilot lots submitted meets the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. All future production lots should be of the same basic granulation.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination and testing.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

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4.4.2.1 Sampling (see 3.1 and 3.2). Each lot of propellant shall be sampled and tested for composition, forms, dimensions and chemical/physical properties in accordance with MIL-STD-652. Each lot of propellant shall be sampled for proving ground tests in accordance with MIL-STD-652 except that ten containers shall be sampled for both initial and subsequent production lots.

4.4.2.2 Permissible individual maximum pressure. If at any time during testing, an individual pressure greater than that specified in 3.2.1 is observed, testing shall cease and the lot shall be rejected (Commander, ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801 shall be notified).

4.4.2.2.1 Differential pressure. If at any time during testing, an individual differential pressure exceeds the applicable requirement, testing shall cease and the lot shall be rejected. If, for any test group, the mean of the differential pressures plus three (3) times the standard deviation exceeds applicable requirement, the lot shall be referred (see 6.12).

4.4.2.3 Charge establishment (see 3.2.2). If the lot does not meet the applicable requirement, testing shall cease and the lot shall be rejected (Commander, ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801 shall be notified).

4.4.2.4 Velocity uniformity (70°F) (see 3.2.3.1). If the product of the factor (0.72) times the velocity standard deviation of the charges tested in accordance with 4.5.1.2 exceeds the applicable requirement, apply referral criteria in accordance with 4.5.1.4. If the referral criteria do not apply the lot shall be rejected.

4.4.2.5 Pressure uniformity

4.4.2.5.1 Average pressure (see 3.2.3.2). If any average pressure obtained during initial or subsequent production at 70°F and 145°F exceeds the applicable requirement, apply referral criteria in accordance with 4.5.1.4. If the referral criteria do not apply the lot shall be rejected

4.5 Test methods and procedures

4.5.1 Proving ground assessments. These tests shall be conducted at a government proving ground in accordance with the applicable acceptance test procedure using M549, RA or PXR-6325 projectiles inert loaded within 0.1 pound of an as fired weight of 96.0 pounds using primer, M82 and fired in cannon M199 with 75 percent (minimum) remaining life. Firing shall be conducted with a

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spindle to charge base stand-off of 1 inch (plus or minus 1/4 inch). Prior to firing, all charges shall be temperature conditioned in a large conditioning box with adequate air circulation at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature. The following tolerances shall apply to all -65°F, 145°F and 70°F conditioning: -65°F from -65°F to -60°F; 145°F from 140°F to 145°F; 70°F from 68°F to 72°F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each charge. Velocity coils will be used to measure the muzzle velocity of each charge. Temperature indicator (thermometer) strips shall be placed on the side (lacing jacket) of all charges or on the side of the metal containers, as applicable, conditioned at 145°F as a further assurance of the proper conditioning temperature. If any indicator strip shows the charge has been exposed to a temperature greater than 145°F, that charge will not be fired.

4.5.1.1 Charge establishment

4.5.1.1.1 Charge slope method. Three or more groups of charges loaded with selected charge weights of propellant in accordance with MIL-C-48243 (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. From the data obtained a charge weight shall be calculated which will yield the prescribed service velocity (see 3.2.2). This calculated charge weight shall then be loaded into charges for the uniformity series (see 4.5.1.2). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established. The resultant will be the recommended charge weight.

4.5.1.1.2 Constant slope method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity curve can be established from test results generated. This curve can than be used as in 4.5.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into charges for the uniformity series and final adjustment is to be made as indicated above for the charge slope method.

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4.5.1.2. Uniformity. Thirty charges shall be assembled with the established charge weight of propellant in accordance with MIL-C-48243. Ten charges each shall be conditioned to -65°F, 70°F and 145°F (as indicated by thermocoupled control charge. Charges fired at 70°F shall be fired alternately with calibration charge and shall be corrected to standard weapon/firing conditions (i.e. the difference between calibration performance under test conditions and standard conditions). Charges fired at -65°F and 145°F shall be intermixed with 5 calibration charges. Firing results at all temperatures shall be corrected to standard projectile weight (see 3.2) and to recommended charge weight conditions (ie. the difference between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. The average pressure-temperature coefficient for the lot shall be calculated from the uncorrected pressure data obtained at 70°F and 145°F.

4.5.1.3 Retest criteria. A lot shall be retested whenever any of the following conditions exist.

a. The final adjustment in charge weight exceeds two percent of the charge weight used in uniformity firings, retest for charge establishment in accordance with 4.5.1.1.

b. If for any reason, the Proving ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested.

4.5.1.4 Referral criteria. A test lot shall be referred to the procuring activity with a copy of test data forwarded to ARRADCOM, ATTN: DRDAR-QAR-R for disposition whenever any of the following conditions exist and whenever the proving ground deems it necessary.

a. Calibration charges fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds 25 feet per second of the expected muzzle velocity.

c. Calibration components are not available.

d. Number of test or calibration charges fired is less than specified.

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e. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, poor projectile flights, etc. Each such event shall be reported in detail on the applicable firing record.

f. Test lot fails velocity standard deviation or average pressure requirement at 70°F and any extreme values is an outlier (see 6.9).

4.5.2 Differential pressure and ignition delay . Two pressure-time gages at different locations (one in the spindle and one at the projectile base) shall be used to record pressure-time traces on all -65°F and +145°F firings. Differential pressure traces shall be obtained by subtracting the projectile gage output from the spindle gage output. The differential pressure of any charge is defined as the maximum amplitude of the first negative wave of the differential pressure trace prior to peak pressure. Any trace which contains no amplitude in the negative direction is considered to have a maximum differential pressure of zero. Ignition delay, the time from primer initiation to a rise in pressure equivalent to ten percent of the maximum expected pressure, shall be recorded on all differential pressure firings.

5. PACKAGING

5.1. Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed Level A.

6. NOTES

6.1 Intended use. The propellant covered by this specification is intended for use in the M203 propelling charge.

6.2 Ordering data. Ordering data shall be in accordance with MIL-A-48078 (PA) with the following exceptions.

6.2.1 Acquisition requirements. Acquisition requirements of MIL-A-48078 (PA) shall apply.

6.2.2 Contract data requirements. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data and one copy of the Propellant Description sheet shall be forwarded to: Commander, ARRADCOM, ATTN: DRDAR-QAR-R and one copy to ATTN: DRDAR-LCA-G, Dover, NJ 07801.

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6.3 Standard deviation. The standard deviation shall be calculated with (N-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.4 Temperature indicator strips. The highest temperature reading from each indicator strip shall be recorded for informational purposes and reported on the applicable firing record.

6.5 Surveillance sample. Samples shall be forwarded to ARRADCOM Dover, NJ, ATTN: DRDAR-LCE for the propllant surveillance test program in accordance with MIL-STD-652.

6.6 Calibration and test charges. Calibration and test charges shall be assembled in accordance with drawing 9281897 and shall be manufactured in accordance with MIL-C-48243 at an applicable load plant.

6.7 Proving ground test summary (see Table I).

6.8 Retest. All original and retest data shall be submitted to ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801.

6.9 Outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_c) is 0.477. Calculate a test value (T_c) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

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TABLE I
Proving Ground Test Summary

Test Phase	Temp °F	Sample Size Cal	Test	Velocity Requirements: Service Velocity fps	Velocity Requirements: Lot Std Dev of Vel fps	Pressure Requirements*: Perm Ind Max Press psi	Pressure Requirements*: Average Pressure Max psi
Charge Establishment	70	3	9	2710	N/A	57,950	47,500
Uniformity	-65	5	10	N/A	N/A	57,950	N/A
	70	10	10	2710	7.0	57,950	47,500
	145	5	10	N/A	N/A	57,950	55,000

*For differential pressure see 4.4.2.2.1 and 3.2.1.1

Where:

X_1 = test result being tested as low outlier
X_2 = test result adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = test result adjacent to the largest test result
X_{10} = Test result being tested as high outlier

X_1 or X_{10} is an outlier if T_t is greater than T_c.

6.10 Test anomalies. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, stickers, poor projectile flights, etc., shall be reported in detail on the applicable firing record.

6.11 Permissible individual maximum pressure (PIMP). The weapon PIMP of 58,800 psi is applicable only when firing with an M549 projectile having the maximum weight allowed by drawing 9281897. Since an M549 or PXR6325 projectile weighing 96.0 pounds is used, the applicable PIMP is calculated to be 57,950 psi.

6.12 Differential pressure. The values given in this specification do not represent a definite boundary between safe and unsafe conditions but serve as triggers to require a safety analysis of the data. If the differential pressure requires referral of the lot, the disposition of the lot shall be referred to US Army Armament Research and Development Command, Attn: DRDAR-QAR-R, Dover, New Jersey 07801.

6.13 International standardization agreements. Certain provisions (see 3.2) of this specification are the subject of international standardization agreement (quadrilateral MOU concerning the standardization of elements of 155MM Howitzer and ammunition, 28 April 78). When amendment, revision or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through Project Manager, Cannon Artillery Weapons Systems/Joint Project Manager, Semi-Active Laser/Guided Projectiles (PM-CAWS/JPM SAL GP).

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6.14 Drawings. Drawings listed in section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.15 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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